

Marcelo HURTADO

BSc Bioengineering – Biotechnological Engineering
PhD student in Computational Biology & Bioinformatics

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EDUCATION

Oct 2023	PhD student  - Paul Sabatier University, Toulouse, France - Cancer Research Center of Toulouse (CRCT) Team: NetB(IO)² Supervisor: Vera Pancaldi Project: Reinforcement learning approaches to control the tumor microenvironment Current achievements: - Develop a multiscale ABM of NLC-CLL differentiation using the Physicell software (Github repository). - Develop a Python and Bash pipeline for Physicell model exploration (including parameter exploration, sensitivity analysis and genetic algorithms) for running in HPC architectures (Github repository). - Apply basic reinforcement learning algorithms into the NLC-CLL model to evaluate cancer cell concentration - Develop an advanced machine learning framework in R to extract clinically relevant features from bulk RNAseq data (pipeML)
Jan 2017 Dec 2023	B.Sc Bioengineering - Mention in Biotechnology - Universidad de Ingenieria y Tecnologia UTEC Lima, Peru. Courses: Synthetic Biology, Biomedical signals, Medical robotics, bioprocesses, biofluids mechanics.
Aug 2021 Dec 2021	Exchange Program Biotechnology and Biomedical Engineering - Instituto de Estudios Superiores Tecnologicos de Monterrey - Monterrey, Mexico. Courses: Genetic engineering, neuroengineering, biomedical imaging

RESEARCH EXPERIENCE

Jan 2021 Aug 2021	Visiting PhD Student Eindhoven University of Technology (TU/e).  Department: Biomedical Engineering Supervisor: Federica Eduati Tasks: Personalisation of mechanistic Boolean models through integration of transcriptomics data. Achievements: Improvement and development of a R package for cell-cell communication from bulk RNA-seq based on kernel and Monte-Carlo algorithms (RaCiNG) .
Apr 2023 Oct 2023	Bioinformatics Engineer Cancer Research Center of Toulouse (CRCT) Team: NetB(IO) Principal Investigator: Vera Pancaldi Tasks: Analisis of transcriptomics data to evaluate patient profiles of early and late stage NSCLC patients. Achievements: Develop a novel framework for characterizing TME patient profiles by constructing transcriptional regulatory networks (TRNs) based on inferred transcription factor (TF) activity and cell type deconvolution from bulk RNA-seq data (CellTFusion). Hurtado M et al. (2024). Transcriptomics profiling of the non-small cell lung cancer microenvironment across disease stages reveals dual immune cell-type behaviors. Front. Immunol. 15:1394965. doi: 10.3389/fimmu.2024.1394965
Jan 2022 March 2022 Aug 2022 Feb 2023	Intern of Bioinformatics Cancer Research Center of Toulouse (CRCT)   Team: NetB(IO)² Principal Investigator: Vera Pancaldi Tasks: Analisis of tumor cell composition on NSCLC patients using computational tools based on transcriptomics data. Achievements: Develop of an algorithm that integrates a combination of multiple first or second-generation deconvolution methods and several cell type signatures to ensure robust and accurate profiling of cell composition starting from bulk RNAseq data (multideconv).

Jan 2021
Aug 2021

Research Assistant | Universidad de Ingenieria y Tecnologia UTEC Lima, Peru.

- **Department:** Bioengineer Department | **Supervisor:** Dr. Alberto Donayre
- **Tasks:** Design of a bioactive polymer based on self-assembling and antimicrobial peptides for wound applications.
- **Achievements:** Propose a genetic cassette for the production of self-assembling peptides with antimicrobial properties via the implementation of genetic engineering techniques and cloning methods (**Gibson assembly, restriction enzymes, golden gate**).

PUBLICATIONS

Hurtado, M., & Pancaldi, V. (2026). CellTFusion: A transcriptional regulatory network framework for the identification of functional multicellular states from bulk RNA-seq data. bioRxiv. <https://doi.org/10.64898/2026.06.30.735682>

Gobbini E., Duplouye P., **Hurtado M.**, ... Pancaldi V., Valladeau-Guilemond J. (2026). Specific dendritic cells spatial organization is associated to ICB Response in Non Small Cell Lung Cancer (NSCLC). bioRxiv. <https://doi.org/10.64898/2026.05.04.720587> (Gobbini E., Duplouye P. and Hurtado M. contributed equally to this work)

Hurtado M., Pancaldi V. (2026). A new pipeline for cross-validation fold-aware machine learning prediction of clinical outcomes addresses hidden data-leakage in omics based predictors. bioRxiv 2026.03.12.711429. <https://doi.org/10.64898/2026.03.12.711429>

Marku, M.*, Chenel, H.*, ..., **Hurtado M.**, et al. Data driven network inference and longitudinal transcriptomics unveil dynamic regulation in Chronic Lymphocytic Leukaemia models. npj Syst Biol Appl (2026). <https://doi.org/10.1038/s41540-025-00645-4>

Bertin A*, Bucher E*, ..., **Hurtado M.**, et al. (2025). PhysiGym: bridging the gap between the Gymnasium reinforcement learning application interface and the PhysiCell agent-based model software. bioRxiv 2025.09.18.677030; doi: <https://doi.org/10.1101/2025.09.18.677030>

Hurtado M., et al. (2025). multideconv - Integrative pipeline for cell type deconvolution from bulk RNAseq using first and second generation methods. bioRxiv. <https://doi.org/10.1101/2025.04.29.651220>

Hurtado M*, Khajavi L*, et al. (2024). Transcriptomics profiling of the non-small cell lung cancer microenvironment across disease stages reveals dual immune cell-type behaviors. Front. Immunol. 15:1394965. doi: 10.3389/fimmu.2024.1394965

ORAL PRESENTATIONS

12/05/2026 14/05/2026	<i>Estimating tumour microenvironment cellular states from bulk RNA-seq reveals biomarkers of clinical outcome in Non Small Cell Lung Cancer (NSCLC) across disease stages</i> EACR - Cancer Genomics, Multiomics and Computational Biology - Essen, Germany.
01/12/2025 04/12/2025	<i>CellTFusion: A novel approach to unravel cell states via cell type deconvolution and TF activity estimated from bulk RNAseq data identifies clinically relevant cell niches</i> Bioinformatics and Computational Biology Conference - Napoli, Italy.
26/11/2025 28/11/2025	<i>Estimating tumour microenvironment cellular states from bulk RNAseq produces biomarkers of clinical outcome across stages</i> 21èmes Journées annuelles du Cancéropôle GSO - Agen, France.
20/07/2025 24/07/2025	<i>CellTFusion: A novel approach to unravel cell states via cell type deconvolution and TF activity estimated from bulk RNAseq data identifies clinically relevant cell niches</i> European Conference on Computational Biology - ISMB/ECCB 2025 - Liverpool, UK.
07/11/2024	<i>"CellTFusion: Transcriptional regulatory networks unravel cell states from immune cell type deconvolution and uncovers cell niches predictive of cancer progression"</i> Journee Bioinfo/Biostat GenoToul - Toulouse, France.
26/11/2023 01/12/2023	<i>"Transcriptional regulatory networks unravel cell states from immune cell type deconvolution and uncovers cell niches predictive of cancer progression"</i> EMBO Workshop Computational models of life: From molecular biology to digital twins Sant Feliu de Guíxols, Spain.

22/11/2023 24/11/2023	<i>"Transcriptional regulatory networks unravel cell states from immune cell type deconvolution and uncovers cell niches predictive of cancer progression"</i> 19th Annual Meeting of the Canceropole GSO - Arcachon, France. ★ ¹
25/09/2023 29/09/2023	<i>"Transcriptional regulatory networks unravel cell states from immune cell type deconvolution and uncovers cell niches predictive of cancer progression"</i> Institut Curie Computational Systems Biology of Cancer 6th edition - Paris, France. ★ ²

POSTER PRESENTATIONS

2025	Estimating NSCLC tumour microenvironment cellular states from bulk RNAseq produces biomarkers of clinical outcome across stages World Conference on Lung Cancer (WCLC) 2025 - Barcelona, Spain.
2025	CellTFusion: A novel approach to unravel cell states via cell type deconvolution and TF activity estimated from bulk RNAseq data identifies cell niches potentially predictive of cancer progression JOBIM 2025 - Bordeaux, France
2024	Transcriptional regulatory networks unravel cell states from immune cell type deconvolution and uncovers cell niches predictive of cancer progression 1st edition Young Scientist Cancer Congress - Toulouse, France
2023	Profiling of the Non-Small Cell Lung Cancer (NSCLC) Microenvironment Across Disease Stages EACR Defence is the Best Attack: Immuno-Oncology Breakthroughs 2023 - Barcelona, Spain.
2023	Profiling of the Non-Small Cell Lung Cancer (NSCLC) Microenvironment Across Disease Stages The Festival of Genomics & Biodata 2023 - London, UK.
2022	Multi-omics Profiling of the Non-Small Cell Lung Cancer (NSCLC) Microenvironment Across Disease Stages and Gender Cancéropôle Grand Sud-Ouest 18th Annual Meeting - Montpellier, France.

REVIEWING AND EDITING ACTIVITIES

- Review Commons
- Clinical Cancer Research
- Bioinformatics

COURSES AND WORKSHOPS

02/12/2024 06/12/2024	Health Data Challenge 2024: Multimodal data integration to quantify tumor heterogeneity in cancer research. Aussois, France.
29/06/2024 05/07/2024	14th Summer school on medicines. University of Sao Paulo, Ribeirao Preto Medical School.
21/04/2024 26/04/2024	Computational Systems Biology for Complex Human Disease: from static to dynamic representations of disease mechanisms. Wellcome Genome Campus, Hinxton, United Kingdom.
26/11/2023 01/12/2023	EMBO Workshop. Computational models of life: From molecular biology to digital twins. Sant Feliu de Guíxols, Spain.
25/09/2023 29/08/2023	6th course on Computational Systems Biology of Cancer: models of data, data for models. Institut Curie, Paris, France.

LEADERSHIP & OUTREACH

Workshop organization – Innovations in immuno-oncology: from data to therapeutic insights

Team leader – iGEM Desing League 2021 Synthetic Biology competition. ★

Project *LECCHAIN*: Improving vaccine thermal tolerance and stability of SARS-CoV-2 antigen using plant lectins.

Press release:

- [Centro Bio](#)
- [Peru21](#)
- [Press coverage](#)
- [ElComercio](#)

SUPERVISION

2026 – Rajath PRABHU MANJESHWAR – **Masters M2 program (6 months)**

Project: Investigating CLL Cell plasticity through Co-Culture systems and Agent Based Modelling.
Cancer Research Center of Toulouse (CRCT)

2026 – Mariajose CAZORLA – **International program (3 months)**

Project: Transcriptomics Profiling of the Tumor Microenvironment in KRAS Mutated Patients with Non-Small Cell Lung Cancer.
Cancer Research Center of Toulouse (CRCT)

2024 – Francesco MASSAINI – **Masters M2 program (6 months)**

Project: Profiling the Immune Landscape: Multi-Omics analysis of the Tumor Microenvironment in Non-Small Cell Lung Cancer.
Cancer Research Center of Toulouse (CRCT)

AWARDS AND ACHIEVEMENTS

- ★ Prix de la Fondation Silab Jean Paufigue
- ★ Best oral presentation award
- ★ Campus France – France Excellence Eiffel PhD scholarship
- ★ Toulouse Foundation Cancer Sante – PhD scholarship
- ★ CARe Graduate School Univ. Paul Sabatier – Doctoral School Fellowship
- ★ CARe Graduate School Univ. Paul Sabatier – M2 internship fellowship
- ★ Research for Peruvian Undergraduates (REPU) program – International stage fellowship
- ★ iGEM Design League – Gold medal, Best Human Practices, Best aligned with SGD
- ★ ERASMUS+ Fellowship

REFERENCES

Dr. Vera **PANCALDI**

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Cancer Research Center of Toulouse
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Dr. Julien **MAZIERES**

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Hospital, University Hospital of Toulouse,
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